Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

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Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

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Outline

1. **Quantum error correction and transversal gates**
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For binary vectors $c, c' \in \mathbb{F}_2^n$, $\mathrm{wt}(c)$ is the number of ones of c, the *overlap* $c \wedge c'$ is the entrywise product, and $c \cdot c' = \sum_i c_i c'_i \bmod 2 = \mathrm{wt}(c \wedge c') \bmod 2$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings*

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act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

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- The weight of $X(a)Z(b)$ is the number of qubits where a or b is 1. An $[\![n, k, d]\!]$ code has dimension 2^k, and d is the minimum weight of a Pauli string that commutes with every stabilizer but is not a stabilizer up to sign.

CSS codes

For binary linear codes $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] is the stabilizer code with stabilizers $X(a)$, $a \in \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_1^\perp$. They commute because $a \cdot b = 0$. Write $\mathcal{C}_2 = \operatorname{rowspan} S$ and $\mathcal{C}_1 = \operatorname{rowspan} \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, with stabilizer rows S $(s \times n)$ and logical rows K $(k \times n)$. The logical basis is

$$|x\rangle_L = 2^{-s/2} \sum_{y \in \mathbb{F}_2^s} |xK \oplus yS\rangle, \qquad x \in \mathbb{F}_2^k.$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
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- $Z(e)$ has syndrome Se^T and is undetected if and only if $e \in \mathcal{C}_2^\perp$. $X(e)$ is undetected if and only if $e \in \mathcal{C}_1$. So Z errors are corrected by decoding $\mathcal{C}_2^\perp$, and X errors by decoding $\mathcal{C}_1$.

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- Logical operators: $X(a)$, $a \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$. With $d_X = \min \operatorname{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2)$ and $d_Z = \min \operatorname{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp)$, the code is $[\![n, k, d]\!]$ with $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$ and $d = \min(d_X, d_Z)$.

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Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \tfrac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3}|x_1 x_2 x_3\rangle.$$

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- H, S and CNOT generate the Clifford group, which is not universal. Adding T or CCZ gives a universal gate set.

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- Magic-state distillation [BK05] uses Clifford operations to turn many noisy non-Clifford states into fewer states with lower error. Each one applies a non-Clifford gate.

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Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode $[15,4,8]$	$T^{\otimes 15}$ acts as logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	A layer of T and $T^\dagger$ acts as logical CCZ.

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CNOT between two blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out transversal T on the Steane code and transversal H on the $[\![15,1,3]\!]$ code.

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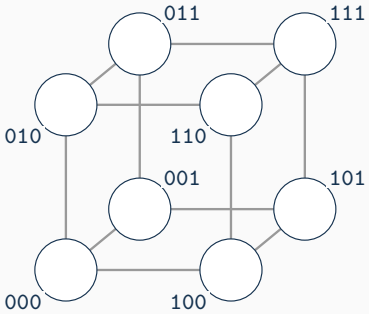
Question: which CSS codes with $k = 3$ have a choice $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ acting as logical CCZ?

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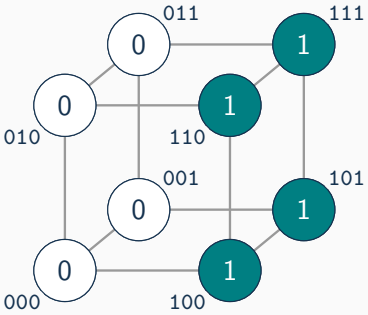
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

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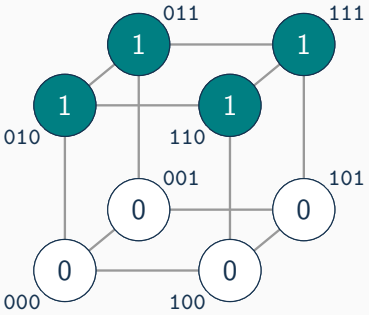
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Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

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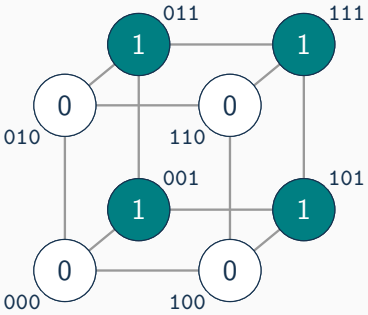
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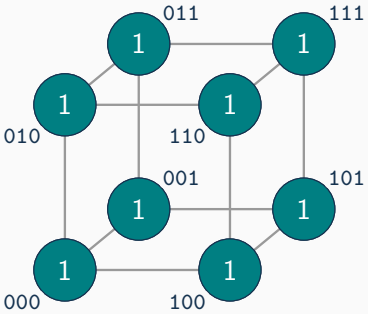
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The constant vector $\mathbf{1}$.

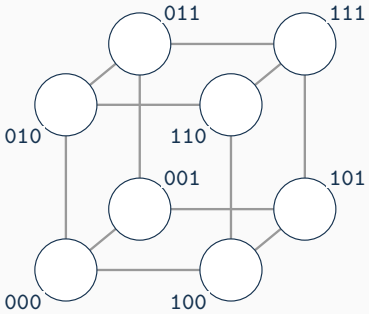
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$\mathrm{RM}(r, m)$ is the code of the vectors given by polynomials of degree at most r. For example, $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

The cube code $[\![8, 3, 2]\!]$

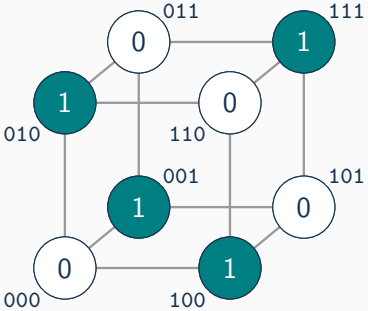


Take $S = \mathbf{1}$, so the only stabilizer generator is $X(\mathbf{1}) = X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

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The codeword $x \cdot u$ for $x = 111$.

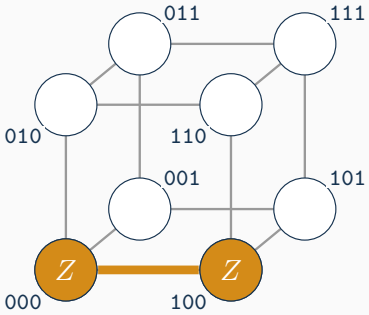
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For $x = 111$ the codeword $x \cdot u$ is $u_1 \oplus u_2 \oplus u_3$.

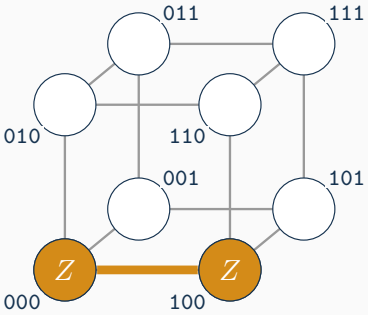
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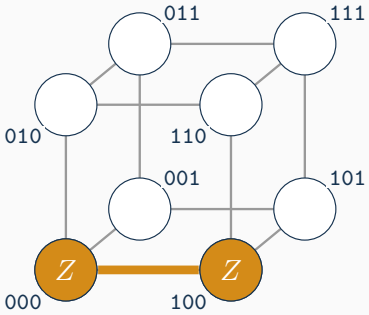


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- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\mathbf{1}_u)$ anticommutes with it, so $d_Z \geq 2$.

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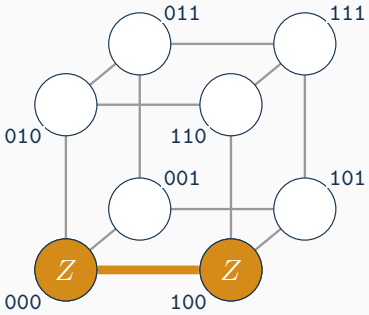


Z on the qubits 000 and 100.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\mathbf{1}_u)$ anticommutes with it, so $d_Z \geq 2$.
- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.

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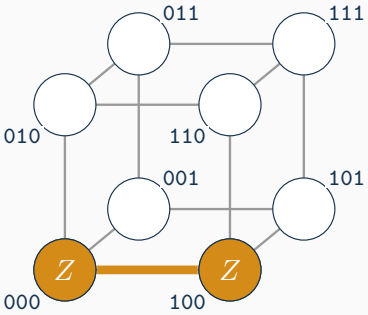


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- $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$.

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This is the smallest interesting colour code: a transversal $T/T^\dagger$ layer gives logical CCZ [CH17].

Codeword weight, physical phase, and logical phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c on T qubits minus the ones on $T^\dagger$ qubits.

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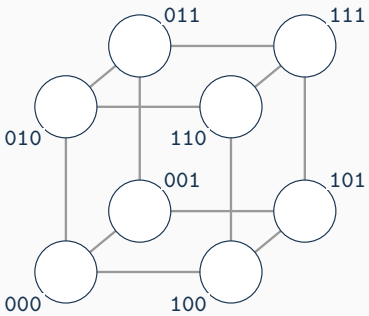
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The product acts as logical CCZ if and only if every codeword of the coset $xK \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

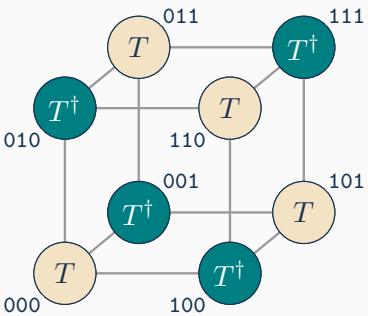
Choosing the signs on the cube



Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

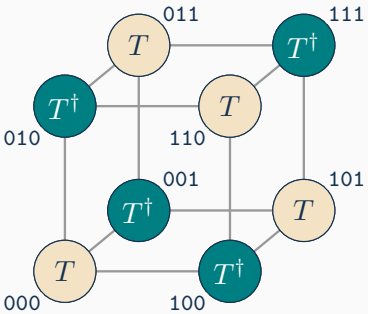
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Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. With the Hamming distance $d_H(p, c) = \mathrm{wt}(p \oplus c)$,

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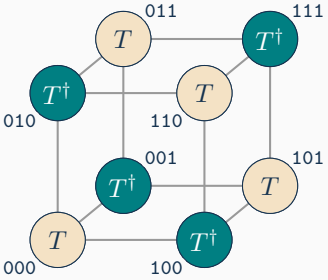
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$x \neq 111$: $p \oplus f_{x,b} = b \oplus (x \oplus 111) \cdot u$ is nonconstant, so $d_H = 4$ and $W_\Gamma = 0$.

$x = 111$: $f_{x,b} \in \{p, \mathbf{1} \oplus p\}$, so $d_H \in \{0, 8\}$ and $W_\Gamma = \pm 4$, phase -1.

Reading the cube phase from overlaps



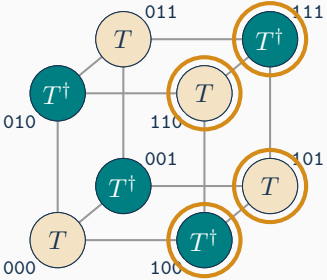
For bits a_1, a_2, a_3, XOR has the integer expansion

$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

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$K_1 = u_1$: the face $u_1 = 1$.

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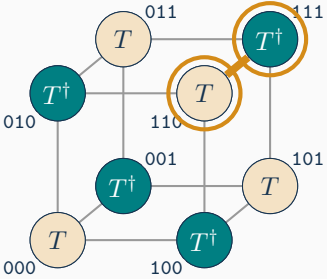
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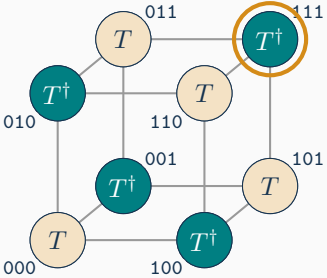
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The vertex 111 is a $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1$ and $W_\Gamma(c) = -4x_1 x_2 x_3 \equiv 4x_1 x_2 x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The same expansion for any number of binary vectors is the *Bravyi–Haah parity identity* [BH12]

$$W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with it and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

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The $[\![8, 3, 2]\!]$ code satisfies all three: rows have $W_\Gamma = 0$, pairs of rows overlap in faces or edges with $W_\Gamma = 0$, and triples overlap in an edge ($W_\Gamma = 0$) or in the vertex 111 ($W_\Gamma = -1$).

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

Outline

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2. The cube code: a worked example of logical CCZ
3. **Triorthogonality and the CCZ conditions**
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ using indicator functions in Reed–Muller codes.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows give the logical X operators. Even-weight rows give the X-type stabilizers.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j \equiv 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every vector in $\mathcal{H}$ has even weight.

Nezami–Haah classify the parent spaces $\mathcal{H}$, then obtain codes by puncturing coordinates. This work fixes S from geometry first, then searches for a compatible K.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17]. They give CCZ up to a diagonal Clifford correction (*quasi-transversal*).

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A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

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Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- $\mathcal{C}_2$ is fixed by projective geometry: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$.
- The logical rows come from a SAT search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- We found codes at block lengths 48 to 496 using S constructed from $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$.

- $\beta_j \neq 0$ for all j: every single Z error is detected.
- $\beta_i \neq \beta_j$ for all $i \neq j$: every pair of Z errors is detected.

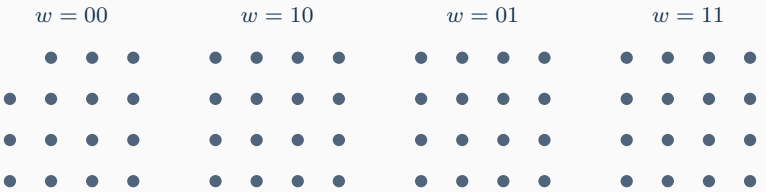
Distinct nonzero columns are points of $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the columns of S alone, for every choice of logical rows K.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

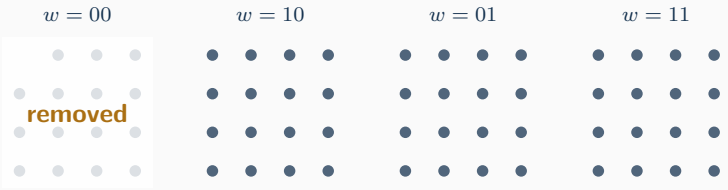
The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



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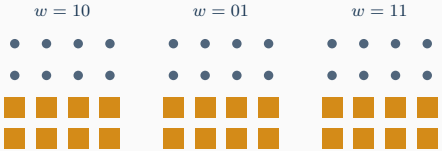


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ code $\mathcal{C}_2$.

Hyperplane sections and codeword weights

Each codeword of $\mathcal{C}_2$ is the linear function $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ evaluated on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



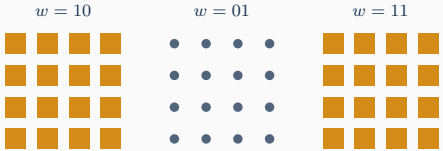
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to point sets that meet every hyperplane in one of two sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective two-weight codes correspond to point sets that meet every hyperplane in one of two sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

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More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p \backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. **Column indicators, logical rows, and phase signs**
6. Lower bounds and open questions

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $\mathrm{wt}(\chi_M)$ is the number of columns of M.

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For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

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Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$, the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

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Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) satisfies these conditions [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

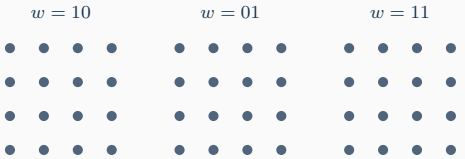
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



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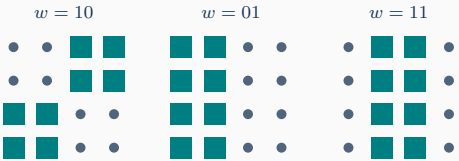
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Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 \oplus u_3$, $1 \oplus u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. Let β_l denote the l-th coordinate function on $\mathcal{P}$ (row l of S). K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.

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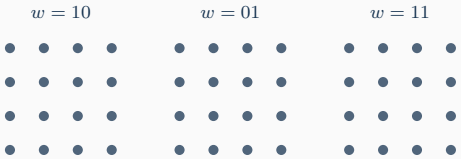
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If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For Q_{48}, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. So Z on these three qubits is a logical operator, and $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\mathrm{wt}(\gamma \wedge c)$, and every codeword $c = xK \oplus yS$ has even weight:

$$\mathrm{wt}(\gamma \wedge c) \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



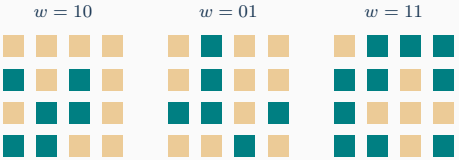
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For Q_{48} [LB26]:

$$\gamma(u, w) = u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),$$
$$Q_1 = u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,$$
$$Q_2 = u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.$$



Teal: $T^\dagger$ $(6, 6, 10$ per block). Gold: T. Total $26\,T + 22\,T^\dagger$.

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Stabilizer supports are Reed–Muller codewords

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 imply

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, where δ_0 is the indicator of the origin and ϵ makes $\mathrm{wt}(c)$ even. Every monomial of degree at least 1 vanishes at $\beta = 0$, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = \mathrm{wt}(c) \bmod 2$, which ϵ makes 0. Hence c is orthogonal to every monomial of degree at most 3, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits logical rows satisfying K1–K3 and KS1–KS3.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no physical $T/T^\dagger$ layer [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat, punctured at the origin. Q_{48} removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with circuit footprint 9.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $Q_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit logical rows K and signs Γ?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public